



Graded Lab 1 Artificial Intelligence

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Instructions

- In this lab assignment, you will solve problem(s) based on what you have learnt in objects and classes.
- There are 4 queries in this assignment.
- **This evaluation instrument, if submitted after due date, will not be evaluated, and a score of zero will be awarded.**
- **Email/paper/other modes of submissions will not be accepted.**
- Upload a pdf **version** of this document.

Due Time: 2 hours after assigning.

Submitting this Assignment

Paste your code and snapshot of output after the question, save the document You will submit (upload) this assignment in LMS moodle. **Paste your code and snapshot of output** after the question, save and upload the document.

Problem:

Basic Python Programming understanding:

1. Write a program to print the **Fibonacci series** up to N terms.
2. Write a program to check whether a number is positive, negative, or zero.
3. Write a Python program to implement the A* search algorithm to find the shortest path from node A to node G.

4. Smart Street Light Agent

Write a Python program to implement a Simple Reflex Agent for a street light. If it is dark, the agent should turn the light **ON**; if it is bright, the agent should turn the light **OFF**. Display the sequence of perceptions and actions.

Query:

Output:



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Query 1:

```
def print_fibonacci(n):
    if n <= 0:
        print("Please enter a positive integer.")
        return []
    a, b = 0, 1
    fib_series = []
    for _ in range(n):
        fib_series.append(a)
        a, b = b, a + b
    return fib_series

n_terms = 10
result = print_fibonacci(n_terms)
print(f"Fibonacci series up to {n_terms} terms:")
print(result)
```

Output:

```
C:\Users\prakh\OneDrive\Desktop\ai class work\workspace> python Problem1.py
Fibonacci series up to 10 terms:
[0, 1, 1, 2, 3, 5, 8, 13, 21, 34]
```

Query 2:

```
def check_number(num):
    if num > 0:
        return "Positive"
    elif num < 0:
        return "Negative"
    else:
        return "Zero"

test_numbers = [15, -7, 0, 42.5, -18.2]
```



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```
print("Number Classification Check:")
print("=" * 35)
for num in test_numbers:
    result = check_number(num)
    print(f"Number: {num:>6} -> {result}")
```

Output:

```
C:\Users\prakh\OneDrive\Desktop\ai class work\workspace> python Problem2.py
```

```
Number Classification Check:
```

```
=====
```

```
Number:    15 -> Positive
Number:    -7 -> Negative
Number:     0 -> Zero
Number:   42.5 -> Positive
Number:  -18.2 -> Negative
```

Query 3:

```
import heapq
```

```
def a_star_search(graph, heuristics, start, goal):
    open_set = []
    heapq.heappush(open_set, (0 + heuristics[start], 0, start, [start]))
    visited = {}

    while open_set:
        f_score, g_score, current, path = heapq.heappop(open_set)

        if current in visited and visited[current] <= g_score:
            continue
        visited[current] = g_score

        if current == goal:
```

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```
        return path, g_score

    for neighbor, weight in graph.get(current, []):
        new_g = g_score + weight
        new_f = new_g + heuristics[neighbor]
        heapq.heappush(open_set, (new_f, new_g, neighbor, path + [neighbor]))

    return None, float('inf')

graph = {
    'A': [('B', 4), ('C', 2)],
    'B': [('D', 5), ('E', 10)],
    'C': [('E', 3), ('F', 8)],
    'D': [('G', 11)],
    'E': [('G', 4)],
    'F': [('G', 5)],
    'G': []
}

heuristics = {
    'A': 10,
    'B': 8,
    'C': 5,
    'D': 7,
    'E': 3,
    'F': 4,
    'G': 0
}

start_node = 'A'
goal_node = 'G'
path, total_cost = a_star_search(graph, heuristics, start_node, goal_node)

print("A* Search Algorithm (Node A to Node G)")
print("=" * 45)
print(f"Start Node          : {start_node}")
```



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```
print(f"Goal Node          : {goal_node}")
print(f"Shortest Path Found: {' -> '.join(path)}")
print(f"Total Path Cost    : {total_cost}")
```

Output:

```
C:\Users\prakh\OneDrive\Desktop\ai class work\workspace> python Problem3.py
```

```
A* Search Algorithm (Node A to Node G)
```

```
=====
```

```
Start Node          : A
Goal Node           : G
Shortest Path Found: A -> C -> E -> G
Total Path Cost     : 9
```

Query 4:

```
def street_light_agent(environment_light):
    if environment_light == "Dark":
        return "Light ON"
    else:
        return "Light OFF"

perceptions = ["Dark", "Dark", "Bright", "Dark", "Bright", "Bright"]
print("Smart Street Light Agent - Simple Reflex Agent")
print("=" * 45)
for i, perception in enumerate(perceptions):
    action = street_light_agent(perception)
    print(f"Step {i+1}: Perception = {perception} -> Action = {action}")

print()
print("Agent Details:")
print("Sensor      : Light Sensor / LDR (detects ambient light)")
print("Actuator     : Street Light Switch (ON/OFF)")
print("Agent Type   : Simple Reflex Agent")
print("Environment: Fully Observable, Single-agent, Static")
```



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Output:

```
C:\Users\prakh\OneDrive\Desktop\ai class work\workspace> python Problem4.py
```

```
Smart Street Light Agent - Simple Reflex Agent
```

```
=====
```

```
Step 1: Perception = Dark -> Action = Light ON
```

```
Step 2: Perception = Dark -> Action = Light ON
```

```
Step 3: Perception = Bright -> Action = Light OFF
```

```
Step 4: Perception = Dark -> Action = Light ON
```

```
Step 5: Perception = Bright -> Action = Light OFF
```

```
Step 6: Perception = Bright -> Action = Light OFF
```

Agent Details:

```
Sensor      : Light Sensor / LDR (detects ambient light)
```

```
Actuator    : Street Light Switch (ON/OFF)
```

```
Agent Type  : Simple Reflex Agent
```

```
Environment: Fully Observable, Single-agent, Static
```